

CHAPTER FOUR

CARBONACEOUS MATERIAL REMOVAL

by

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1. INTRODUCTION

Basically all aerobic biological treatment processes operate on the same principles: the various treatment processes i.e. trickling filters, aerated lagoons, contact-stabilization, extended aeration, etc. differ only in the conditions under which the biological reactions are constrained to operate. The activated sludge process comprises the flow régime in the reactor, the sizes, number and configuration of the reactors, recycle flows, influent flow and other features incorporated either deliberately, or present inadvertently or unavoidably. Whereas the response of the organisms is in accordance with their nature, that of the process is governed by both the organism characteristics and the physical features that define the process.

The ecology of the suspended sludge mass in the reactors of the activated sludge processes is complex comprising numerous species of bacteria and many species of higher organisms that prey on the bacteria. Despite the complex nature of the ecosystem, it would appear that the kinetic behaviour of the process, insofar as it reflects the energy removal from the wastewater, can be modelled as if the system were governed by an equivalent or surrogate bacterial mass, having characteristics derived in part from the observed response of the treatment processes. It must be emphasised that this surrogate mass has characteristics very different from those of a pure bacterial culture. This point is mentioned because criticism of this approach has often hinged on the incompatibility of the behaviour of the equivalent bacterial mass with that observed in a pure bacterial culture.

In this chapter the basic kinetics of the carbonaceous material removal, and its incorporation in the process equations are briefly stated, and application to design is set out in detail. The design equations presented are for the steady state completely mixed single reactor activated sludge process. The steady state solution is of prime importance because (1) it forms the basis for preliminary design of any activated sludge process be it aerobic, anoxic-aerobic or anaerobic-anoxic-aerobic; (2) the relationships also are useful in indicating operational control procedures for the process.

If certain minimum conditions are imposed on the process, nitrification will occur. With nitrification the oxygen demand can show a considerable increase, of up to 40% more than that required for carbonaceous material oxidation. Nitrification is not dealt with in this chapter but is considered in detail in Chapter 5.

2. BIOLOGICAL KINETICS

When a heterotrophic organism population under aerobic conditions is brought into contact with biodegradable organic material* consisting of soluble readily biodegradable and particulate slowly biodegradable COD fractions (see Chapter 2), its response may be described qualitatively as follows:

- (1) The soluble readily biodegradable COD passes directly through the cell wall and is metabolized at a high rate.
- (2) The particulate slowly biodegradable COD is adsorbed onto the organisms and storage of COD takes place. This reaction is very rapid and effectively removes all the particulate and colloidal COD from the wastewater. The stored COD is broken down by extracellular enzymes and transferred through the cell wall and metabolized in the same manner as the readily biodegradable COD fraction in (1) above. The rate of enzymatic breakdown is relatively slow and constitutes the limiting rate in the overall synthesis reaction, at about one tenth of the rate for the readily biodegradable COD.
- (3) A fraction of the COD metabolized is converted into new cell material, the balance is utilized to generate energy to bring about the conversion and eventually, lost as heat. The energy is generated by supplying an electron acceptor, oxygen, so that the oxygen utilized is directly related to the COD lost. There is strong experimental evidence that the fraction of COD converted to metabolic material is constant with respect to the COD utilized, Payne (1971). This fraction is called the specific growth yield coefficient, Y_{COD} (mg COD/mg COD). Usually the coefficient is expressed in terms of the volatile solids formed per unit of COD utilized, Y_h , (mg VSS/mg COD) and is related to Y_{COD} by Eq. 1.6, i.e. $Y_h = Y_{\text{COD}}/f_{\text{cv}}$.
- (4) Concomitantly with (3) above, but distinct from it, there is a *net* loss of live mass called endogenous mass loss. Not all the live mass that disappears *per se* is lost as energy, a fraction remains (17 to 20 per cent, McKinney and Symons, 1964) as unbiodegradable organic residue, called endogenous residue. The mass of oxygen to be supplied for en-

*Also called substrate or biodegradable COD.

ogenous mass loss is proportional to the volatile mass that disappears from the system. This *net* behaviour is believed to be the outcome of a more complex set of mechanisms involving the growth of primary organisms (e.g. bacteria) and predators on these organisms (e.g. protozoa). Dold, Ekama and Marais (1980) proposed and formulated the net behaviour as a uniform organism mass having a death-regeneration component, an approach which approximates the prey-predator system under certain circumstances.

Using the death-regeneration approach Dold *et al* (1980) in their development of the general activated sludge process model, and van Haandel and Marais (1981) in their extension of this model to denitrification, have shown that the response of all the known process types (including nutrient removal processes) can be simulated very closely, even under cyclic flow and load conditions. Under constant flow and load, the death-regeneration approach reduces to the endogenous mass loss approach. This latter approach leads to relatively simple process equations (Marais and Ekama, 1976) and constitutes the basic design equations in this monograph, see Eqs. 4.6 to 4.19.

3. PROCESS KINETICS

3.1 Mixing régime

In the activated sludge process, the mixing régime in the reactor and the sludge return influence the response of the process – therefore consideration must be given to reactor kinetics.

There are two extremes of mixing; completely mixed and plug flow.

In the completely mixed régime the influent is instantaneously and thoroughly mixed (theoretically) with the reactor contents. Hence the effluent flow from the reactor has the same constitution as the reactor contents. The reactor effluent flow passes to a settling tank; the overflow from the tank is the stabilized waste stream, the underflow is concentrated mixed liquor and recycled back to the reactor.

In the completely mixed system the rate of return of the underflow has no effect on the reactor process except if an undue sludge build-up occurs in the settling tank. The shape of the reactor is approximately square or circular in plan, and mixing is usually by mechanical aerators but can also be effected by bubble aeration. Examples are extended aeration plants, aerated lagoons, Pasveer ditch and completely mixed activated sludge plants.

In a plug flow régime, the reactor usually is a long channel type basin. The influent is introduced at one end of the channel, flows along the channel axis and is mixed by air spargers set along one side of the channel. Theoretically each volume element of liquid along the axis is assumed to remain unmixed with the elements leading and following. Discharge to the settling tank takes place at the other end of the channel. To inoculate the influent waste flow with organisms, the underflow from the settling tank is returned to the in-

fluent end of the channel. This creates an intermediate flow régime deviating from true plug flow conditions depending on the magnitude of the recycled underflow. Conventional activated sludge plants are of the intermediate flow régime type with recycle ratio varying from 0.25 to 3 times the influent flow rate. If the recycle rate is raised very high, the mixing régime approaches that of complete mixing.

Intermediate flow régimes are also achieved by having completely mixed reactors in series, or by step-aeration. In the latter, the influent is fed at a series of points along the axis of the plug flow type reactor. Both configurations require, for inoculation purposes, recycling of the sludge from the settler to the influent point.

The mean kinetic response of an activated sludge process, i.e. sludge mass, daily sludge production, daily oxygen demand and effluent quality is adequately, indeed accurately, given by assuming the process is completely mixed and the influent flow and load are constant. This allows the plant volume, the mass of sludge wasted daily and mean daily oxygen demand to be determined by relatively simple formulations. Peak oxygen demands which arise under cyclic flow and load conditions can be estimated subsequently quite accurately by applying an adjustment to the mean daily demand, so also for other parameters. These adjustments have been developed from simulation studies, using the general model of Dold, Ekama and Marais (1980), on processes operated under cyclic and under constant flow and load conditions. For details see Chapter 6, Section 7.6.2.

3.2 Sludge age

A diagrammatic sketch of the completed mixed process with sludge recycle is given in Figure 4.1. This sketch differs from that usually presented in that the sludge wasted each day is abstracted directly from the reactor – the common practice is that the waste sludge is abstracted from the secondary settler underflow. Sludge abstraction directly from the reactor leads to a method of control of the sludge age, called the *hydraulic control of sludge age*, which has significant advantages compared to control by wastage via the underflow, see Section 10.

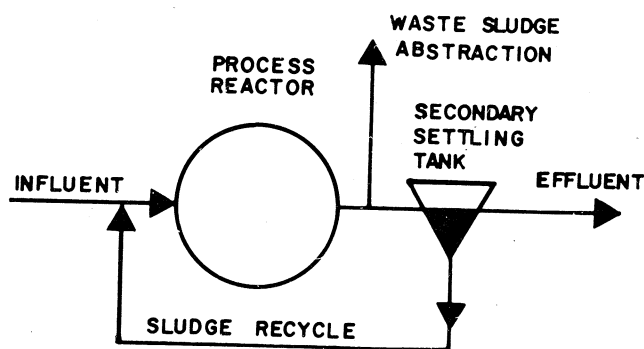


FIGURE 4.1: A diagrammatic representation of the completely mixed activated sludge process with hydraulic control of sludge age and sludge recycle.

The sludge age, R_s , in days, is defined by:

$$R_s = \frac{\text{mass of sludge in reactor}}{\text{mass of sludge wasted per day}} \text{ (days)} \quad (4.1)$$

By abstracting the sludge directly from the reactor, the wasted liquor and reactor liquor concentrations are the same. If a sludge age of, say, 10 days is required, one tenth of the *volume of the reactor* must be wasted every day.* This can be achieved by a constant draw-off at flow rate, q , per day where $q \cdot 1 \text{ day} = v$, the volume to be wasted, hence

$$R_s = \frac{XV_p}{Xq} = \frac{V_p}{q} \quad (4.2)$$

where

V_p = volume of the process reactor (ℓ)

3.3 Nominal hydraulic retention time

In activated sludge theory the volume of the process per unit of volume of influent flow is known as the nominal process hydraulic retention time i.e.

$$R_{hn} = V_p/Q \quad (4.3)$$

where

R_{hn} = average nominal process hydraulic retention time (d)

Q = daily average influent flow rate (ℓ/d)

3.4 Effluent COD concentration

Under normal activated sludge process operating conditions in South Africa, where the sludge ages are in excess of 10 days (to ensure nitrification *and* nutrient removal) the nature of the influent COD in municipal wastewaters is such that the COD concentration in the effluent is inconsequential in the process design – the soluble readily biodegradable COD fraction is completely utilized in a very short period of time (less than 1 hour) and the particulate COD, whether biodegradable or unbiodegradable, is adsorbed or enmeshed in the sludge

flocs, and settles out with the sludge in the secondary settling tanks. Consequently, *the effluent COD concentration is comprised virtually wholly of the soluble unbiodegradable COD* (from the influent) plus the COD of the sludge particles which escape with the effluent due to imperfect operation of the secondary settling tank (see Chapter 3, Section 3.3). Hence the effluent COD concentration, S_{te} , is approximately given by

$$S_{te} = S_{us} \quad (4.4a)$$

for filtered samples, or,

$$S_{te} = S_{us} + f_{cv} \cdot X_{ve} \quad (4.4b)$$

for unfiltered samples, where

S_{us} = unbiodegradable COD in the effluent
 $= S_{usi} = f_{us} S_{ti}$ (mgCOD/ ℓ) (see Eq. 2.3)

X_{ve} = volatile solids concentration in the effluent (mgVSS/ ℓ)

f_{cv} = COD/VSS ratio of the volatile solids
 $= 1.48 \text{ mgCOD/mgVSS}$.

4. PROCESS DESIGN EQUATIONS

Once it is recognized that all the COD in the influent, except the soluble unbiodegradable COD, is either utilized by the micro-organisms to form micro-organism mass, or remains in the process and accumulates as inert sludge mass, it follows that the mass of sludge produced and the carbonaceous oxygen demand in the process are functions of the mass of COD to be treated daily; the greater the daily COD mass load, the greater the sludge production and carbonaceous oxygen demand.

The equations** given below give the masses of sludge generated in the process and the oxygen demand for COD removal as a function of the total COD load, the wastewater characteristics i.e. the unbiodegradable soluble and particulate COD fraction (f_{us} and f_{up}) and the sludge age. The constants in the equations i.e. the specific yield coefficient Y_h , the specific endogenous mass loss rate b_h , the unbiodegradable fraction of the active mass f , and the COD/VSS ratio of the sludge f_{cv} , as well as their temperature dependencies are given in Table 4.1.

TABLE 4.1 KINETIC CONSTANTS AND THEIR TEMPERATURE DEPENDENCY FOR THE STEADY STATE CARBONACOUS DEGRADATION ACTIVATED SLUDGE MODEL (ex MARAIS AND EKAMA, 1976)

Constant	Symbol	Temperature Dependency Equation	θ	Standard Value (20°C)	Eq. No.
Yield coefficient (mg VSS/mg COD)	Y_h	remains constant	1,000	0,45	(4,5)
Endogenous respiration rate (/d)	b_h	$b_{hT} = b_{h20}\theta^{(T-20)}$	1,029	0,24	
Endogenous residue (mg VSS/mg VSS)	f	remains constant	—	0,20	
COD/VSS ratio (mg COD/mg VSS)	f_{cv}	remains constant	—	1,48	

*This assumes that the mass of sludge in the secondary settling tanks is negligible with respect to that in the process reactors. This assumption is valid when the process is operated at high underflow recycle ratios ($\cong 1:1$) and when the sludge age is longer than about 3 days, (see Section 10 below).

**The derivation of these equations is given by Marais and Ekama (1976) and a list of symbols used in the equations is given in Appendix 1.

In the equations below, the prefix M denotes the mass of the parameter which follows in brackets.

(1) Influent

$$M(S_{ti}) = Q \cdot S_{ti} \quad \text{mg COD/d} \quad (4.6)$$

$$M(S_{bi}) = Q \cdot S_{bi} \quad \text{mg COD/d} \quad (4.7a)$$

$$= Q \cdot S_{ti} (1 - f_{us} - f_{up}) \quad \text{mg COD/d} \quad (4.7b)$$

$$= M(S_{ti}) (1 - f_{us} - f_{up}) \quad \text{mg COD/d} \quad (4.7c)$$

$$M(X_{ii}) = Q \cdot X_{ii} \quad \text{mg VSS/d} \quad (4.8a)$$

$$= Q \cdot f_{up} S_{ti} / f_{cv} \quad \text{mg VSS/d} \quad (4.8b)$$

$$= M(S_{ti}) f_{up} / f_{cv} \quad \text{mg VSS/d} \quad (4.8c)$$

(2) Process

$$M(X_a) = V_p X_a \quad \text{mg VSS} \quad (4.9a)$$

$$M(X_e) = V_p X_e \quad \text{mg VSS} \quad (4.9b)$$

$$M(X_i) = V_p X_i \quad \text{mg VSS} \quad (4.9c)$$

$$M(X_v) = V_p X_v \quad \text{mg VSS} \quad (4.9d)$$

$$M(O_c) = V_p O_c \quad \text{mg O/d} \quad (4.9e)$$

(a) The Active Volatile Solids Mass (mg VSS)

$$\begin{aligned} M(X_a) &= M(S_{bi}) Y_h R_s / (1 + b_h R_s) \quad (\text{mg}) \\ &= (1 - f_{us} - f_{up}) M(S_{ti}) Y_h R_s / (1 + b_h R_s) \end{aligned} \quad (4.10)$$

(b) The Endogenous Residue Volatile Solids Mass (mg VSS)

$$M(X_e) = f b_h R_s M(X_a) \quad (\text{mg}) \quad (4.11)$$

(c) The Inert Volatile Solids Mass (mg VSS)

$$\begin{aligned} M(X_i) &= M(X_{ii}) R_s \quad (\text{mg}) \\ &= M(S_{ti}) (f_{up} / f_{cv}) R_s \end{aligned} \quad (4.12)$$

(d) The Total Volatile Suspended Solids Mass (mg VSS)

$$\begin{aligned} M(X_v) &= M(X_a) + M(X_e) + M(X_i) \\ &= \frac{Y_h R_s M(S_{bi})}{(1 + b_h R_s)} (1 + f b_h R_s) + M(X_{ii}) R_s \quad (\text{mg}) \\ &= M(S_{ti}) R_s \left\{ \frac{(1 - f_{us} - f_{up}) Y_h}{(1 + b_h R_s)} (1 + f b_h R_s) + \frac{f_{up}}{f_{cv}} \right\} \end{aligned} \quad (4.13)$$

(e) The Total Suspended Solids Mass (mg TSS)

$$M(X_t) = M(X_v) / f_i \quad (4.14)$$

where

f_i = MLVSS/MLSS ratio of the sludge.

(f) The Carbonaceous Oxygen Demand (mg O/d)

$$\begin{aligned} M(O_c) &= M(O \text{ synthesis}) + M(O \text{ endogenous mass loss}) \\ &= (1 - f_{cv} Y_h) M(S_{bi}) + f_{cv} (1 - f) b_h M(X_a) \quad (\text{mg O/d}) \\ &= M(S_{ti}) (1 - f_{us} - f_{up}) \left\{ (1 - f_{cv} Y_h) \right. \\ &\quad \left. + f_{cv} (1 - f) b_h \frac{Y_h R_s}{(1 + b_h R_s)} \right\} \end{aligned} \quad (4.15)$$

Knowing the mass of mixed liquor (MX_t or MX_v) in the process, the volume of the process is determined from the value specified for the MLSS or MLVSS concentration, X_t , or X_v respectively,

$$V_p = M(X_t) / X_t = M(X_v) / X_v \quad (4.16)$$

Knowing the volume V_p , the total nominal hydraulic retention time, R_{hn} , then is found from the specified daily average flow rate Q , using Eq. 4.3.

The above design equations lead to the following important conclusions:

The mass of volatile solids in the reactor is a function only of the mass of COD utilized per day and the sludge age. Consequently, insofar as the mass of sludge is affected, it is immaterial whether the mass of COD utilized arises from a low daily flow with a high COD concentration, or a high daily flow with a low COD concentration; provided $M(S_{ti})$ is the same in both instances, the masses of sludge will be identical. However, the hydraulic retention times will differ, being long in the first and short in the second instance, respectively. The hydraulic retention time, therefore, is incidental to the COD mass utilized, the MLVSS and the daily flow – it serves no basic kinetic function in steady state operation.

The active volatile mass $M(X_a)$ in the process is the live organism mass which performs the biodegradation processes. The remaining volatile masses, $M(X_e)$ and $M(X_i)$ are inactive and do not serve any function insofar as the biodegradation mechanisms in the process are concerned. The active fraction of the sludge mass with respect to the volatile solids f_{av} is given by

$$f_{av} = M(X_a) / M(X_v) \quad (4.17)$$

substituting for $M(X_a)$ and $M(X_v)$ and noting that

$$M(S_{bi}) = (1 - f_{us} - f_{up}) M(S_{ti}) \text{ and}$$

$$M(X_{ii}) = (f_{up} / f_{cv}) \cdot M(S_{ti}) \text{ yields}$$

$$\frac{1}{f_{av}} = 1 + f b_h R_s + \frac{f_{up} (1 + b_h R_s)}{f_{cv} Y_h (1 - f_{us} - f_{up})} \quad (4.18)$$

If the total suspended solids mass (TSS) is used as the basis for determining the active fraction, the active fraction of the sludge mass with respect to the total suspended solids f_{at} is given by

$$f_{at} = f_i f_{av} \quad (4.19)$$

where

f_i = MLVSS/MLSS ratio of the sludge.

The design equations set out above form the starting point for all the activated sludge processes considered in this manuscript, from the relatively simple single reactor completely mixed aerobic process, to the more complex multi-reactor anoxic-aerobic and the anaerobic-anoxic processes. For these more complex systems the basic equations apply only if the constraints imposed on the processes are adhered to. Provided this is done the effects of nitrification and nitrification-denitrification and the associated oxygen demands can be formulated as additional to the basic equations. That this "simplified" approach is adequate for design has been established by the close correlation achieved when comparing the mean response of the process predicted by the general theory (and verified experimentally) with that predicted by the basic and additional equations.

It is important to take cognizance of the constraints. For example, in anoxic-aerobic processes an upper limit is imposed on the fractional mass of sludge that may exist unaerated. If exceeded, then *inter alia* the mass of sludge produced per day increases and the carbonaceous oxygen demand decreases below that predicted by the basic equations. The reason for these deviations lies in the kinetics of degradation of the slowly biodegradable particulate material – if the aerobic fraction of the sludge mass is too small the particulate material is only partially metabolized and accumulates in the process as additional volatile solids; concomitantly the carbonaceous oxygen demand is reduced. Evidently where such conditions apply the basic equations become inappropriate; however approximate solutions are sometimes obtainable by simulation using the general model, (see Arkley and Marais (1982) and Sehayek and Marais (1981) for a detailed discussion).

5. STEADY STATE DESIGN CHART

Plots of Eqs. 4.10 to 4.15 and 4.18 to 4.19 are shown in Figure 4.2 for a unit mass of $M(S_{ii})$. The values of the kinetic constants, Y_h , b_h , f_{cv} and f are those listed in Table 4.1 for 20°C and have been verified in extensive laboratory and pilot scale investigations (Ekama and Marais, 1978). The values of the unbiodegradable particulate and soluble COD fractions (f_{up} and f_{us} respectively) are listed in Table 4.2 and are for raw and settled municipal wastewaters.

The plots in effect give the masses $M(X_a)$, $M(X_e)$, $M(X_i)$, $M(X_v)$ and $M(X_t)$ contained in the reactor for a unit mass of COD applied per day for sludge ages up to 30 days. Also shown is the associated mass of oxygen to be supplied per day for carbonaceous material degradation and the active fraction of the sludge with respect to VSS (X_v) and TSS (X_t). The volumes of the reactors at two different sludge ages will be in direct proportion to the masses $M(X_t)$ or $M(X_v)$ if the mixed liquor concentrations, X_t or X_v are specified to be the same in both plants.

TABLE 4.2 INFLUENT WASTEWATER FRACTIONS FOR SETTLED AND UNSETTLED SEWAGE

Sewage Fraction	Sewage Unsettled	Sewage Settled
Soluble unbiodegradable fraction, f_{us} (mg COD/mg COD)	0,05	0,08
Particulate unbiodegradable fraction, f_{up} (mg COD/mg COD)	0,13	0,04
MLVSS/MLSS ratio (f_i)	0,75	0,83

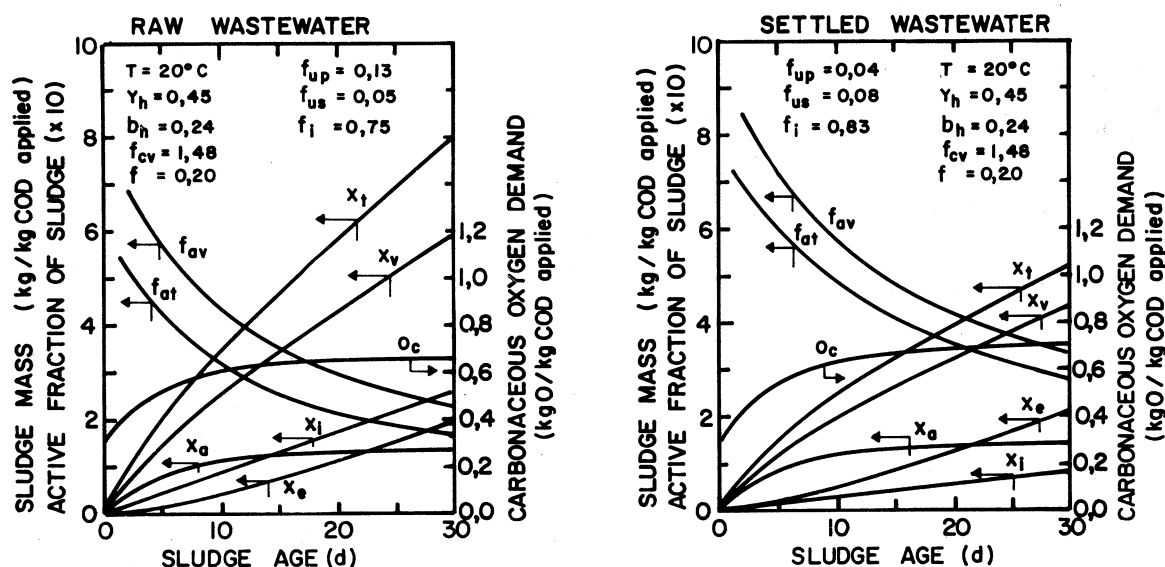


FIGURE 4.2: Active (X_a), endogenous (X_e), inert (X_i), volatile (X_v), total (X_t), and carbonaceous oxygen demand (O_c) per kg COD applied to the biological process, and active fraction with respect to volatile solids (f_{av}) and total solids (f_{at}) versus sludge age for raw (Figure 4.2a, left) and settled (Figure 4.2b, right) wastewaters at 20°C

The diagrams show that the active mass increases fairly rapidly with an increase in sludge age up to about 10 days sludge age, whereafter it increases only marginally. The carbonaceous oxygen demand shows a similar behaviour; at zero sludge age the oxygen demand is that due to synthesis, additional oxygen demand at sludge ages greater than zero is due to endogenous respiration which is proportional to the active mass. In contrast the fractions of endogenous and inert solids increase rapidly relative to the active mass at sludge ages greater than 10 days and consequently in processes treating raw wastewater at sludge ages longer than 10 days, only a small fraction of the sludge mass is active.

In design, the following sequence of calculation is suggested:

Select values for f_{up} and f_{us} which are believed to best represent these fractions of the wastewater. Select the MLVSS/MLSS ratio of the sludge produced. Determine:

- (a) S_{upi} (Eq. 2.4), S_{usi} (Eq. 2.3), X_{ij} (Eq. 2.6)
- (b) $M(S_{ti})$ and/or $M(S_{bi})$
- (c) $M(X_a)$ (Eq. 4.10), $M(X_e)$ (Eq. 4.11), $M(X_i)$ (Eq. 4.12), $M(X_v)$ (Eq. 4.13), $M(O_c)$ (Eq. 4.15) and $M(X_t)$ (Eq. 4.14)
- (d) V_p (Eq. 4.16)
- (e) R_{hn} (Eq. 4.3)
- (f) S_{te} (Eqs. 4.4a or b)

In the design sequence set out above the input COD and its characteristics will be governed by the specific waste flow. The parameter that requires selection is the sludge age; this will depend on the objectives for the plant, for COD removal only, nitrification etc. Specification of the sludge age, therefore, is an important design decision and requires special consideration, see Section 11. For our general purpose however, we will proceed to show how the various parameters such as reactor volume, oxygen demand, etc. are affected by the sludge age.

6. PROCESS VOLUME REQUIREMENTS

In Section 4 above a procedure is set out whereby the volume of the reactor is estimated. This involves determining the mass of sludge accumulated for a specified sludge age and a specified COD mass loading per day; the volume is then determined by diluting the mass of sludge accumulated to a specified mixed liquor concentration. Once the volume is fixed, the retention time, or aeration time, also is fixed (by Eq. 4.3). Hence retention time, or aeration time, is immaterial in the design and it is merely a consequence of the kinetics governing the process. This point is mentioned specifically because some design procedures lay stress on the aeration time as a design parameter, a procedure that can result in serious miscalculation of the volume requirements.

Compare, for example, two plants operating at the same sludge age, both receiving the same mass of COD per day but the first at high concentration and low flow and the second at a low concentration and high flow. If designed on a specified retention time, the volume of the first will be much smaller than that of the second whereas the sludge mass contained in the reactors will be identical. Consequently the first plant may have an inordinately high MLSS concentration which may cause problems in the secondary settling tank.

Retention time therefore is a completely inappropriate parameter for design and other purposes such as a criterion for comparing the tankage requirements of different plants.

To form an overall impression of the volume requirement and in order to make valid comparisons on the volume requirements between different plants, the volume per unit COD input is a much superior parameter. This method involves the specification of an MLVSS or MLSS concentration (X_v or X_t) from which the reactor volume is determined from the sludge mass produced per unit COD load at the required sludge age (see Figure 4.2). Furthermore, in order to assess the effects of primary sedimentation it is preferable to relate the volume requirements to the input COD to the treatment works; for example for settled wastewater, the COD load on the process is reduced i.e. $S_{ti(process)} = S_{ti(works)} (1 - f_{rps})$ where f_{rps} = fraction of COD removed by primary sedimentation. It is $S_{ti(process)}$ which fixes the sludge mass and because for settled wastewater S_{ti} is up to about 40% less than for raw wastewater, the process volume for settled wastewater accordingly will be much lower than that for raw wastewater.

From Eqs. 4.13, 4.14 and 4.16 the process volume requirements on the basis of process COD load is found by calculating the mass of sludge, $M(X_t)$, generated from the total influent COD load $M(S_{ti})$ and assuming an MLVSS/MLSS ratio for the sludge mass i.e.

$$V_p X_t f_i = Q \cdot S_{ti} R_s \left\{ \frac{(1 - f_{us} - f_{up}) Y_h (1 + f_{b_h} R_s)}{(1 + b_h R_s)} + \frac{f_{up}}{f_{cv}} \right\} \quad (4.20)$$

and hence the process volume requirements per unit COD load to the works for a specified MLSS concentration X_t and assumed MLVSS/MLSS ratio f_i is given by

$$\frac{V_p}{Q \cdot S_{ti}} = \frac{R_s}{X_t f_i} \left\{ \frac{(1 - f_{us} - f_{up}) Y_h (1 + f_{b_h} R_s)}{(1 + b_h R_s)} + \frac{f_{up}}{f_{cv}} \right\} \quad (4.21a)$$

Taking the wastewater characteristics for raw wastewater given in Table 4.2, and the values of the kinetic constants given in Table 4.1, a plot of process volume requirements in m^3/kg COD applied per day versus sludge age for different MLSS concentrations in kg/m^3 is given in Figure 4.3. Clearly, the lower the specified MLSS concentration and the longer the sludge age, the larger the process volume per unit COD load.

The process volume requirements may also be determined from the equivalent COD load per capita. The vertical axis on the right hand side of Figure 4.3 is

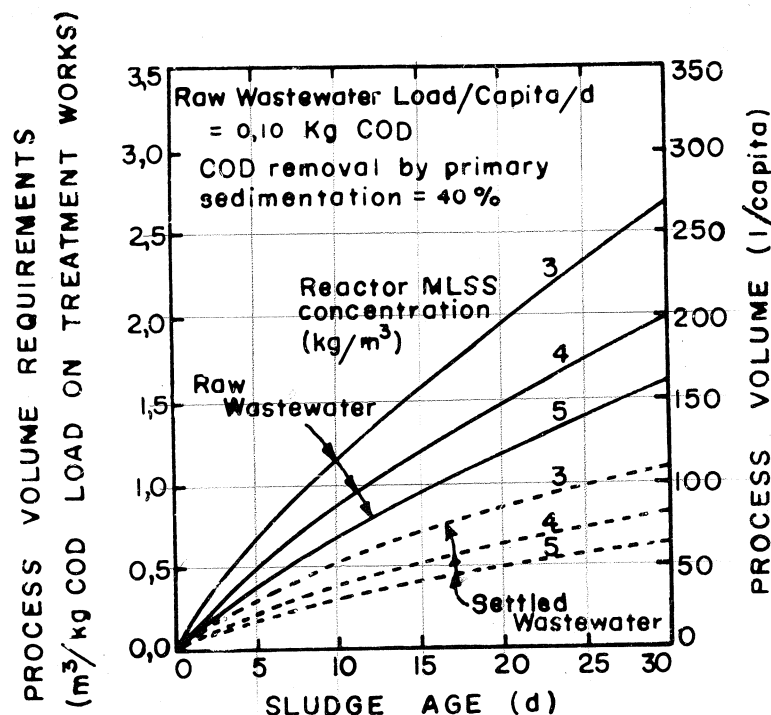


FIGURE 4.3: Process volume requirements in m^3/kg COD raw wastewater load versus sludge age at different average process MLSS concentrations for raw and settled wastewater (assuming 40% COD removal by primary sedimentation). Process volume requirements in l/capita also given based on a raw wastewater COD contribution of $0,10 \text{ kg COD/capita/d}$.

scaled on the basis that the equivalent COD load per capita for raw wastewater is $0,10 \text{ kg COD/d}$. Hence treating raw wastewater at a sludge age of 25 days and an MLSS concentration of 5 kg/m^3 , a process volume of 145 l per capita is required or $1,45 \text{ m}^3/\text{kg COD}$ applied per day.

The comparative process volume requirements for settled wastewater per unit COD load on the treatment works also can be determined from Eq. 4.21a, provided the COD fraction removed by primary sedimentation is incorporated in Eq. 4.21a by multiplying right-hand side of Eq. 4.21a by $(1 - f_{\text{rps}})$ and inserting the appropriate f_{up} and f_{us} values for settled wastewater i.e.

$$\frac{V_p}{S_{\text{ti}}Q} = \frac{R_s}{X_t f_i} \left\{ \frac{(1 - f_{\text{us}} - f_{\text{up}}) Y_h (1 + f_{\text{b}} R_s)}{(1 + b_h R_s)} + \frac{f_{\text{up}}}{f_{\text{cv}}} \right\} (1 - f_{\text{rps}}) \quad (4.21b)$$

where S_{ti} = total COD concentration to treatment works.

The process volume requirements for settled wastewater are shown also in Figure 4.3 for the f_{up} , f_{us} and f_i values given in Table 4.2 assuming primary sedimentation removes 40% of the COD load ($f_{\text{rps}} = 0,40$). From Figure 4.3, treating settled wastewater at a sludge age of 25 days and an MLSS concentration of 5 kg/m^3 , a process volume of $0,60 \text{ m}^3/\text{kg}$ raw COD load to the works or 60 l per capita is required. Comparing the process volume treating raw or settled wastewaters it can be seen that a significant reduction in process volume can be obtained through primary sedimentation.

7. CARBONACEOUS OXYGEN DEMAND

The mean daily carbonaceous oxygen demand per unit COD discharged to the process is shown plotted versus sludge age at 20°C for raw and settled wastewater in Figures 4.2a and 4.2b respectively (calculated from Eq. 4.15) – for treatment of raw wastewater $S_{\text{ti}} = \text{COD of the raw wastewater}$; for treatment of settled wastewater $S_{\text{ti}} = (\text{COD of raw wastewater})(1 - f_{\text{rps}})$. These diagrams show that for sludge ages longer than 15 days the increase in carbonaceous oxygen demand per unit process COD load $\{M(\text{O}_c)/M(S_{\text{ti}})\}$ is marginal with increase in sludge age for both raw and settled wastewater. The carbonaceous oxygen demand per unit process COD load for raw and settled wastewater is usually within 10% of each other, with the demand for settled wastewater being the higher value. This is because compared to raw wastewater, more of the total COD in settled wastewater is biodegradable, i.e. $(1 - f_{\text{us}} - f_{\text{up}})$ usually is greater for settled wastewater than for raw wastewater (see Eq. 2.7) but the difference is small. This can be seen in Figures 4.2a and 4.2b – at 20 days sludge age, the carbonaceous oxygen demand per unit COD load is $0,635 \text{ kgO/kgCOD}$ for raw wastewater and $0,682 \text{ kgO/kgCOD}$ for settled wastewater. Although there is only a small difference in carbonaceous oxygen demand *per unit process COD load* between raw and settled wastewaters there is a substantial difference in the O_2 demand per unit COD load to the works. For settled wastewater, this is given by $0,682 (1 - f_{\text{rps}})$ which for $f_{\text{rps}} = 0,40$ gives $0,41 \text{ kgO/kgCOD}$ load on the works.

Clearly primary sedimentation will lead to significant aeration energy savings – because primary settling tanks remove about 30 to 40% of the raw influent COD, the carbonaceous oxygen demand for settled wastewater will be about 30 to 40% lower than that for raw wastewater.

The carbonaceous oxygen demand is the oxygen demand for the oxidation of the influent COD only. In nutrient removal processes oxygen is also required for nitrification i.e. the oxidation of ammonia to nitrate, and some oxygen is recovered during denitrification i.e. reduction of nitrate to nitrogen gas. The total oxygen demand for a nutrient removal process therefore is given by the sum of the carbonaceous and nitrification demands less that recovered by denitrification. The procedures for calculating the oxygen demand for nitrification and the oxygen recovered by denitrification are discussed in Chapters 5 and 6 respectively.

Owing to the daily cyclic nature of the COD load the carbonaceous oxygen demand will vary concomitantly. The TKN load will also vary daily cyclically; in nitrification-denitrification plants the oxygen demand for nitrification and oxygen recovery with denitrification conjointly with the carbonaceous demand give rise to a net cyclic total demand. From simulations using the general model of Dold *et al* empirical rules for estimating the daily peak and trough net oxygen demands have been developed and are discussed in Chapter 6, Section 7.6.2.

8. DAILY SLUDGE PRODUCTION

The mass of sludge produced per day by the activated sludge process is equal to the mass of sludge wasted per day and is called waste activated sludge or secondary sludge. From the definition of sludge age (see Eq. 4.1), the mass of sludge produced per day $M(\Delta X_t)$ is given by the mass of sludge in the process $\{M(X_t)\}$ divided by the sludge age (R_s) i.e.

$$M(\Delta X_t) = M(X_t)/R_s \quad \text{mg TSS/d}$$

Substituting Eq. 4.14 for $M(X_t)$ and simplifying, yields the sludge produced per day per mg COD load on the biological process i.e.

$$\frac{M(\Delta X_t)}{M(S_{ti})} = \frac{1}{f_i} \left\{ \frac{(1 - f_{up} - f_{us}) Y_h}{(1 + b_h R_s)} (1 + f_{b_h} R_s) + \frac{f_{up}}{f_{cv}} \right\} \quad (4.22)$$

A plot of the daily total sludge mass (TSS) produced per unit COD load on the biological process (Eq. 4.22) versus sludge age (R_s) is shown in Figure 4.4 for raw and settled wastewater (see Table 4.2 for wastewater characteristics). From Figure 4.4, it can be seen that the mass of sludge produced in the activated sludge process (per unit COD load on the biological process) decreases as the sludge age increases for both raw and settled wastewater but the rate of decrease is negligible at sludge ages longer than 20 days. Treating settled wastewater results in lower secondary sludge production per unit COD load on the biological process than treating raw wastewater. This is because the unbiodegradable particulate COD fraction (f_{up}) in settled wastewater is lower than that in raw wastewater.

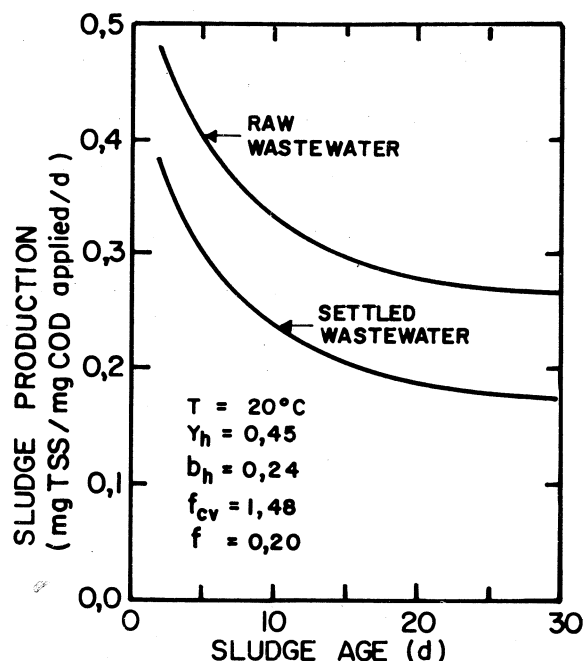


FIGURE 4.4: Daily total suspended solids (TSS) sludge production in the activated sludge process (secondary sludge) per unit COD load on the process versus sludge age for raw ($f_{up}=0.13$, $f_{us}=0.05$ and $f_i=0.75$) and settled ($f_{up}=0.04$, $f_{us}=0.08$ and $f_i=0.83$) wastewaters at 20°C

Temperature effects on secondary sludge production are small – sludge production at 14°C* is about 5% greater than at 20°C, a difference which is completely masked by the uncertainty in the estimates of the wastewater characteristics f_{up} and f_{us} and the MLVSS/MLSS ratio (f_i) of the sludge.

Although the sludge production of a plant treating settled wastewater is lower than that of a plant treating raw wastewater, the *total* sludge mass of sludge to be disposed of for a plant treating settled wastewater is higher because the total sludge includes both primary and secondary sludge – in plants treating raw wastewater, only secondary sludge is generated. The *total* sludge produced treating settled wastewater is slightly higher (by 10 to 20%) than that treating raw wastewater – this is because some of the settleable solids are biodegradable and are oxidized in the activated sludge process.

The daily production of secondary and primary sludges (when primary settling tanks are included in the treatment works) is the mass of sludge that needs to be disposed of by sludge handling methods. Sludge handling and disposal is not dealt with in this manual but the design of nutrient removal activated sludge processes should not be seen as separate from the design of the sludge disposal unit processes, in fact all unit processes of the treatment works from raw water pumping to ultimate disposal of sludge, should be viewed as an integrated system where the design of one unit process depends on the unit processes before it, and decisions

*Sludge production at 14°C can be calculated by adjusting b_h for temperature using Eq. (4.5) in Table 4.1.

on its design may affect the design of unit processes following it. The following examples will illustrate this – (1) in sludge handling practice there is evidence that a mixture of primary and secondary sludges thickens and dewater better than secondary sludge alone; however, in biological nutrient removal practice it has been found that when treating raw wastewater, better removals of nitrogen and phosphorus are obtained than when treating settled wastewater – ease of sludge thickening and dewatering therefore need to be weighed against the higher nutrient removal; (2) when phosphorus-rich waste biological sludges are subjected to anaerobic conditions, the phosphorus is released from the solid phase to the liquid phase; consequently, *supernatant* from waste sludge gravity thickeners may be phosphate rich and should not be discharged to the activated sludge process – in design consideration should be given to chemically treating the gravity thickener supernatant to precipitate the phosphorus or, thickening the sludge by dissolved air flotation to prevent phosphorus release; (3) to improve biological excess P removal, the Johannesburg Municipality specifically operates one of the anaerobic digesters in an acid condition for the production of short chain organic compounds which are then discharged to the anaerobic reactor of the activated sludge process (see Chapter 3, Section 3.3) – consideration must be given at the design stage to the desirability of providing a specially designed facility for this purpose.

9. NUTRIENT REQUIREMENTS FOR SLUDGE PRODUCTION

All live biological material and many inert organic materials contain nitrogen and phosphorus. In the volatile suspended solids in the activated sludge process, the nitrogen content (as N with respect to VSS, f_n) ranges between 9% and 12% with an average of about 10%, and the phosphorus content (as P with respect to VSS, f_p), under purely aerobic conditions, ranges between 1% and 3% with an average of about 2,5%.

Under daily average conditions, the mass of nitrogen incorporated into the sludge mass is equal to the nitrogen content of the daily waste sludge. Accepting that the nitrogen content of the active, endogenous and inert volatile fractions are equal (which in most instances is a reasonable assumption), and equating the *concentration* of nitrogen required for daily sludge production per litre influent, N_s , to the nitrogen content of the waste sludge yields

$$Q \cdot N_s = f_n M(X_v)/R_s$$

Substituting for $M(X_v)$ from Eq. 4.10 and solving for the nitrogen requirements per unit COD, N_s/S_{ti} yields

$$\frac{N_s}{S_{ti}} = f_n \left[\frac{Y_h(1-f_{us}-f_{up})}{(1+b_h R_s)} (1 + f_{b_h} R_s) + \frac{f_{up}}{f_{cv}} \right] \quad (4.23)$$

where

$$\begin{aligned} f_n &= \text{nitrogen content of the volatile solids} \\ &= 0,10 \text{ mg N/mg VSS} \end{aligned}$$

To estimate the P requirement for daily sludge production, it is not correct to assume that the P content of the active, endogenous and inert volatile fractions is the same. The P content of the endogenous and inert volatile fractions (f_p) remain approximately constant at 1,5% whereas the P content of the active mass (γ) can vary from 3% to as high as 35% depending on the different conditions in the process. Indeed, it is the propensity of the active mass to accumulate large quantities of P that is exploited in biological P removal processes. Accepting the different P contents of the active fraction and the endogenous and inert fractions, the concentration of P per ℓ influent (P_s) required for daily sludge production per mg/ ℓ COD load (S_{ti}) is given by

$$\frac{P_s}{S_{ti}} = \left\{ \frac{(1-f_{up}-f_{us})Y_h}{(1+b_h R_s)} (\gamma + f_p f_{b_h} R_s) + f_p \frac{f_{up}}{f_{cv}} \right\} \quad (4.24)$$

where

$$\begin{aligned} \gamma &= \text{P content of the active mass (mg P/mgVASS)} \\ f_p &= \text{P content of the inert and endogenous mass} \\ &\quad (\text{mg P/mg VSS}) = 0,015 \end{aligned}$$

The P incorporated into the sludge mass per unit COD (P_s/S_{ti}) is also the P removal from the wastewater per unit COD, $\Delta P/S_{ti}$. In the activated sludge process, under purely aerobic conditions the γ coefficient is about 0,03 mgP/mgVASS: when an anoxic reactor is included in the process, the coefficient is increased to about 0,06 mgP/mgVASS; when an anaerobic reactor is incorporated in the process, then, depending, amongst others, on the readily biodegradable COD concentration in the anaerobic reactor, the γ coefficient can vary between 0,06 and 0,35 mgP/mgVASS (see Chapter 7, Section 3.3). By stimulating high γ coefficients, with the aid of the anaerobic reactor, high P removals may be effected. This is the basis of biological P removal and is discussed in detail in Chapter 7.

A plot of Eqs. 4.23 and 4.24 is given in Figure 4.5 for $f_n = 0,10$ mg N/mg VSS, $\gamma = 0,03$ mgP/mgVASS, $f_p = 0,015$ mgP/mgVSS, $Y_h = 0,45$ mgVSS/mgCOD, $b_h = 0,24/\text{d}$ at 20°C , $f_{cv} = 1,48$ mgCOD/mgVSS, and $f = 0,20$, versus sludge age (R_s) for raw ($f_{us} = 0,05$ and $f_{up} = 0,13$) and settled ($f_{us} = 0,08$ and $f_{up} = 0,04$) wastewaters.

From Figure 4.5, it is evident that greater quantities of nitrogen and phosphorus are required for sludge production from raw than from settled wastewaters. This is because greater quantities of sludge are produced per mgCOD process load when treating raw wastewaters. Furthermore, the nutrient requirements decrease as the sludge age increases because net sludge production decreases as sludge age increases. Generally, for sludge ages greater than 10 days, the nitrogen removal attributable to net sludge production is less than 0,025 mgN/mgCOD applied. As the influent TKN/COD ratio for domestic wastewater is in the approximate range 0,07 to 0,12, only a minor fraction of the influent nitrogen is removed by sludge wastage; additional nitrogen removal is obtained only by nitrification and denitrification, the kinetics of which are discussed in Chapters 5 and 6. With regard to P removal, as the total P/COD ratio of domestic wastewaters usually ranges

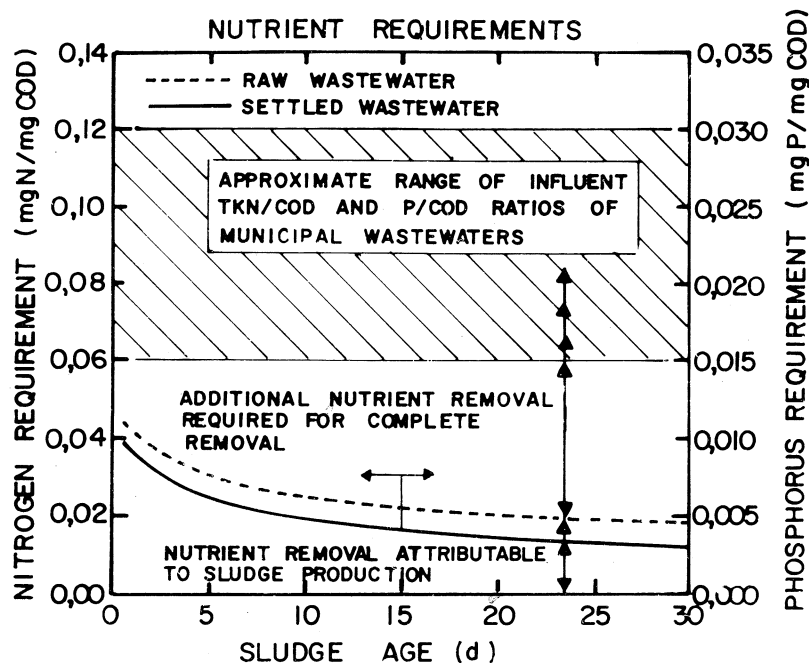


FIGURE 4.5: Normal nutrient nitrogen and phosphorus requirements per kgCOD load in the activated sludge process versus sludge age together with influent nutrient loads per kgCOD.

between 0,015 to 0,030, only a minor fraction of phosphorus, 0,005 mgP/mgVSS, can be removed by the aerobic activated sludge process; excess biological P removal is obtained by stimulating high γ coefficients in processes by incorporating an anaerobic reactor. Details of excess biological P removal are considered in Chapter 7.

10. PROCESS DESIGN AND CONTROL

The parameter of fundamental importance in the design and control of the activated sludge process is the sludge age. It is instructive to consider how this parameter influences both the control and design of the process and to contrast it to other design and control parameters that have been implemented in the past.

10.1 Sludge age versus load factor as control parameter

A widely accepted basis for design (especially in North America and Europe) is the Load Factor (LF). It is also known as the Sludge Loading Rate (SLR) and has been used principally when the BOD_5 forms the influent energy measurement parameter. The objective of the load factor was to form an estimate of the Food/Micro-organism (F/M) ratio and as a result the load factor was defined as

$$LF = \frac{\text{mass of BOD load/d}}{\text{mass of sludge in process}} = \frac{M(S_{BOD})}{M(X_v) \text{ or } M(X_t)} \quad (4.25)$$

where

$M(S_{BOD})$ = mass of BOD load per day (mg)

$M(X_v), (X_t)$ = mass of volatile (mgVSS) or total (mgTSS) suspended solids in process respectively.

A difficulty with the approach is that both the $M(X_v)$ and $M(X_t)$ measurements include unbiodegradable and inert material. To represent the F/M ratio correctly, the LF should be defined as follows:

$$LF = F/M = M(\Delta S_{BOD})/M(X_a) \quad (4.26)$$

where

$M(\Delta S_{BOD})$ = mass change in BOD across plant per day (mgBOD/d)

$M(X_a)$ = active mass in process (mgVASS)

As a practical control parameter, the F/M ratio as defined by Eq. 4.26 is virtually unusable principally because $M(X_a)$ cannot be measured directly. It is for this reason that the LF was *approximated* by Eq. 4.25* with $M(X_v)$ or $M(X_t)$ as the solids parameter. In this form, however, the LF is an uncertain parameter against which to correlate the response of the process because $M(X_v)$ depends on both the biodegradable COD and the inert particulate material in the influent. For example, for the same biodegradable COD concentration, the LF will differ between settled and raw wastewaters whereas the oxygen demand and $M(X_a)$ will be the same for both. Furthermore the LF will be different depending on whether the settleable solids are measured in terms of MLVSS or MLSS. When the COD forms the influent energy measurement parameter, the LF is defined by

*Because the effluent BOD usually is very small with respect to the influent BOD, the change in BOD mass load per day $M(\Delta S_{BOD})$ is closely approximated by the BOD mass load per day $M(S_{BOD})$.

Eq. 4.27 which is even further out of place because the COD measure includes the unbiodegradable soluble and particulate COD fractions

$$LF = \frac{M(S_{COD})}{M(X_v) \text{ or } M(X_t)} \quad (4.27)$$

where

$M(S_{COD})$ = mass of COD load per day.

If Eq. 4.27 is written in terms of the mass change of COD per day across the plant i.e. $M(\Delta S_{COD})$, then account of only the soluble unbiodegradable COD is taken. The unbiodegradable particulate COD affects the LF by increasing both the $M(\Delta S_{COD})$ and $M(X_v)$ or $M(X_t)$ so that different load factors can be obtained depending on the unbiodegradable particulate COD fraction.

To overcome the problem of the unbiodegradable COD fractions on the LF, Marais and Ekama (1976) defined the F/M ratio in terms of COD as the mass change of biodegradable COD per day per unit active volatile mass and, to distinguish this parameter from the LF, called it the Substrate Utilization Rate with respect to the active mass, SUR_a i.e.

$$SUR_a = F/M = M(\Delta S_{COD})/M(X_a) \quad (4.28)$$

where

$M(\Delta S_{COD})$ = mass change in biodegradable COD per day.

Again the SUR_a parameter is of little practical value because neither the $M(\Delta S_b)$ (due to the unknown unbiodegradable particulate COD content of the wastewater) nor the $M(X_a)$ can be measured. However, using Eq. 4.28 as a basis, Marais and Ekama (1976) showed that the Substrate Utilization Rates with respect to the active mass (SUR_a), volatile mass (SUR_v) and total mass (SUR_t) can all be written in terms of sludge age, but different SUR 's are obtained for a fixed sludge age depending on the parameters for measuring the sludge concentration (X_a , X_v or X_t) and the influent energy (COD or BOD) (see Figure 4.6).^{*} Similarly for the same sludge age, different values of the LF or SLR can be obtained depending on the measurement parameters for the sludge concentration and influent energy.

The dependency of the SUR 's, LF 's or SLR 's on the measurement parameters for the sludge concentration and influent energy is undesirable because it provides a permanent source of error and misunderstanding at, and between all levels of technical staff concerned with the process. But perhaps of greater importance is the fact that *all the different SUR 's, LF 's or SLR 's are functionally related to the sludge age – no additional information can be derived from the SUR 's, LF 's or SLR 's that cannot be derived from the sludge age.*

The sludge age therefore can replace completely the LF, SLR and SUR as a reference parameter. Furthermore, the sludge age can be fixed by a simple process control procedure if the process is appropriately designed. This control procedure is much more practical than procedures based on the LF approach.

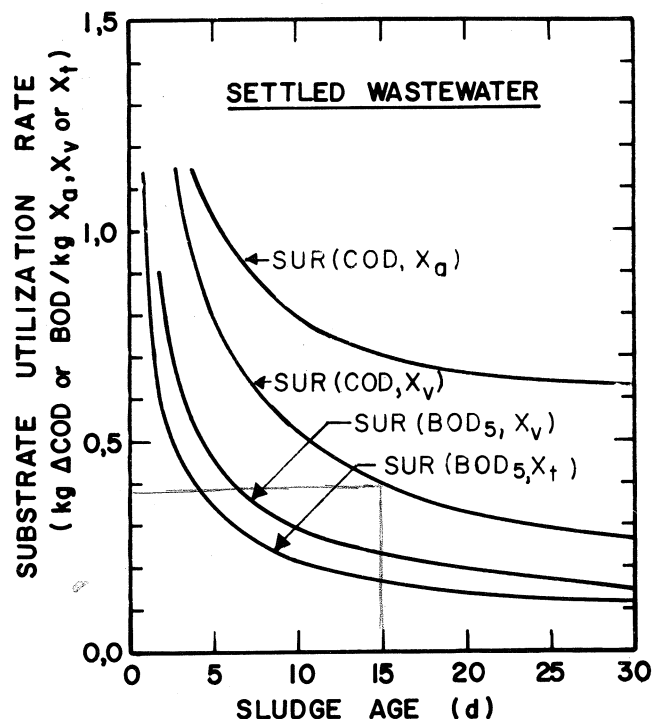


FIGURE 4.6: Comparison between various substrate utilization rates (SUR) and sludge age.

10.2 Process control

The usual procedure of the activated sludge process involves keeping the sludge concentration at some value either specified from design considerations or established from experience on the plant behaviour. Knowing the sludge concentration (X_v or X_t), to check that the LF is within the desired limits requires extensive testing throughout the day of the influent COD (or BOD) concentration and flow pattern to determine the daily COD (or BOD) mass load. To calculate the corresponding sludge age, a history of the daily sludge mass wastage needs to be kept. Often in conventional plants, the waste sludge is abstracted from the secondary settling tank underflow to benefit from the thickening function of the secondary settlers. However, the sludge concentration of the underflow varies considerably with the daily cyclic flow through the plant (see Figures 4.7 and 4.8). Hence, each time sludge is wasted it is necessary to measure the underflow concentration and calculate the sludge age.

From the above, it is clear that in order to know the sludge age or the LF accurately, intensive testing of the plant and influent is required. This is manageable on large plants where the technical supervision is adequate, but on small plants both the Load Factor and sludge age usually are unknown.

The discussion above demonstrates that wasting sludge from the secondary settler underflow does not result in a simple and accurate control of either the sludge age or the LF. However, by using a different sludge wasting procedure, an accurate and simple con-

^{*}The derivation of some of these SUR values is given by Marais and Ekama and those for well settled wastewater (i.e. $f_{up} = 0,000$) is shown in Figure 4.6. It should be noted that if volatile inert material is present in the influent i.e. $f_{up} > 0,00$, e.g. in partially settled and raw wastewaters, a set of SUR values can be calculated for each selected value of f_{up} .

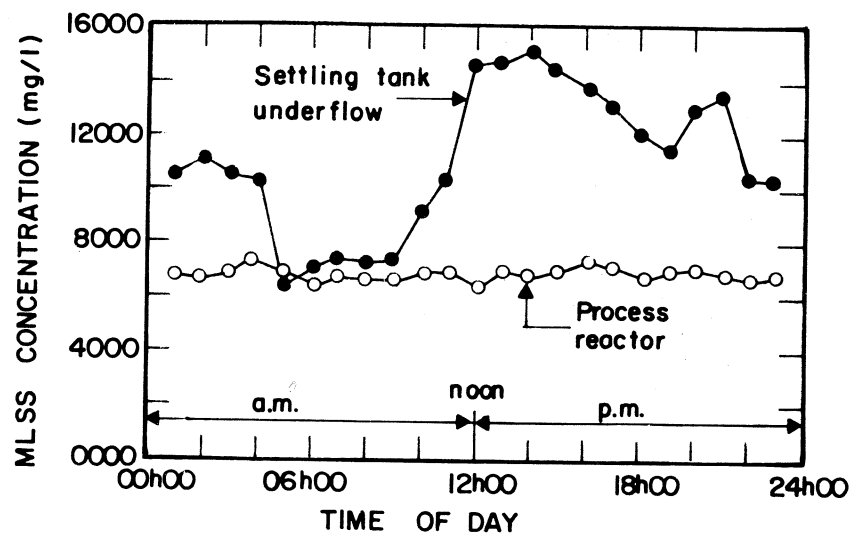
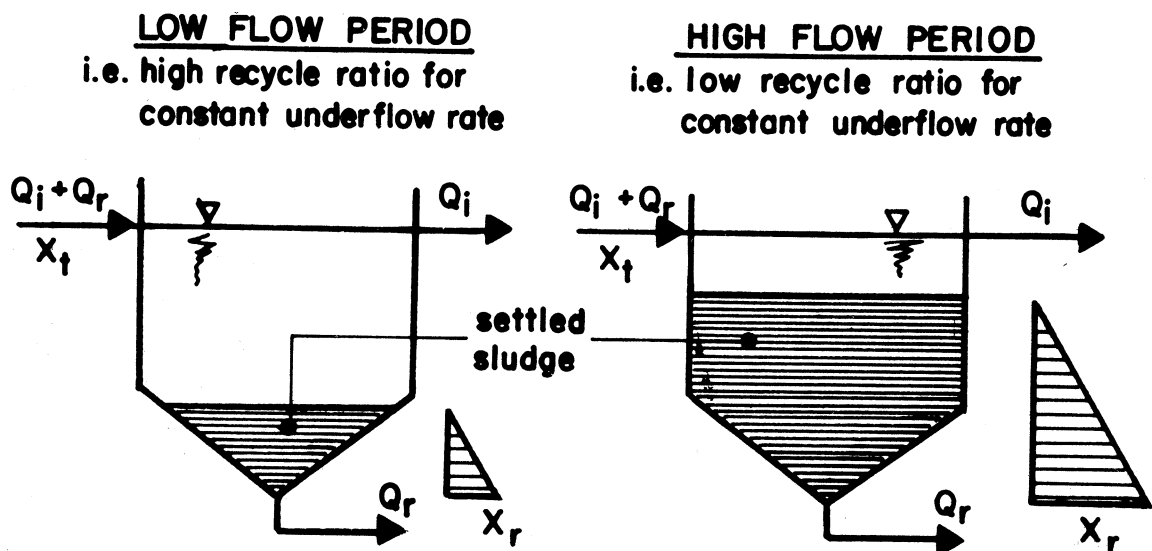


FIGURE 4.7: Experimental data from a full scale activated sludge plant illustrating stability of the reactor MLSS concentration compared with that in the secondary settler underflow.



Q_i = INFLUENT FLOW RATE (m^3/d)

Q_r = UNDERFLOW RATE (m^3/d)

X_t = SLUDGE DENSITY IN REACTOR (mg MLSS/l)

X_r = SLUDGE DENSITY IN UNDERFLOW (mg MLSS/l)

$$X_r = \frac{(Q_i + Q_r)}{Q_r} \cdot X_t = \frac{s+1}{s} \cdot X_t$$

$s = Q_r / Q_i$ = UNDERFLOW RECYCLE RATIO

FIGURE 4.8: Increased sludge accumulation and higher underflow sludge concentration for (i) decreased recycle flow for fixed influent flow and (ii) increased influent flow for fixed recycle flow.

tol of the sludge age is possible. This different sludge wastage procedure is called the hydraulic control of the sludge age.

10.3 Hydraulic control of sludge age

Hydraulic control of sludge age was first proposed and implemented in a generalized form by Garrett in 1958, based on a method of "modified sewage aeration" implemented by Setter, Carpenter and Winslow (1945).^{*} It operates as follows: Suppose a sludge age of 10 days is specified. A satellite settling tank or dissolved air flotation unit is provided, completely independent of the secondary settling tank, to which 1/10 of the reactor volume is pumped every day. The supernatant is discharged to the effluent channel (or pumped back to the reactor),^{**} the thickened sludge is pumped to the digesters or drying beds. That this method gives the correct sludge age can be seen from Eq. 4.2 and the procedure works because the mixed liquor concentration changes insignificantly over the day. This is verified by comparing the stability of the concentration of mixed liquor in the reactor and the concentration of sludge in the underflow over the day in one of the major plants of Johannesburg, Figure 4.7 (Nicholls, 1975).

Controlling the sludge age in small works may require pumping at regular times, every day or every few days, provided the volume pumped is such that the required volume, v , per day is abstracted. On large plants the satellite settling tank (or flotation unit) may be designed for continuous operation receiving a constant flow, or, operated over a part of the day.

An important point about hydraulic control of the sludge age is that *irrespective of the flow through the plant* if a fixed fraction of the volume of the reactor is removed and wasted every day, the sludge age is fixed. If the COD mass load per day on the plant remains constant, the sludge concentration will remain constant automatically. If the COD mass load increases, the sludge concentration will increase automatically, to maintain the same sludge age. Thus, by monitoring the MLSS concentration and its changes for a fixed sludge age an indirect measure is obtained of the COD load and load change.

By means of the hydraulic control procedure, the sludge age may be changed by simply changing the volume wasted per day. If say, the sludge age is reduced from 25 days to 20 days by hydraulic control, the full effect of the change will become apparent only after about 15 days. Thus the organisms have an opportunity to adapt gradually to the change in load.

Hydraulic control of sludge age is particularly relevant to plants with sludge ages longer than about 4 days because for these plants the mass of sludge contained in the secondary settling tanks is a minor fraction of the total mass of sludge in the system. At sludge ages shorter than 4 days the mass of sludge in the secondary settling tanks becomes appreciable with respect to the

total mass of sludge in the system. Hydraulic control will have to take cognizance of this and accuracy of the control will require additional testing. It should be noted that irrespective of the sludge age of the plant, the satellite settling tank required for hydraulic control will always be very small with respect to the secondary settling tanks – at 2 days sludge age the retention time is usually about 4 to 6 hours and hence the satellite settling tank has to accommodate 1/12 to 1/8 of the daily average flow; at 25 days sludge age the retention time is about $\frac{3}{4}$ to $1\frac{1}{4}$ days and the satellite settling tank has to accommodate about 1/20 of the daily average flow; in contrast the secondary settling tank has to accommodate 2 to 4 times the daily average flow.

With regard to design, hydraulic control of the sludge age has the following implications: It devolves a greater onus on the designer and removes responsibility from the plant operator: It becomes essential that the designer calculates the sludge mass more exactly, to provide sufficient reactor volume under the design load to allow for the required concentration of MLSS in the reactor at the specified sludge age. Also, the settling tank design, recycle and aeration capacities must be adequate. If these aspects are catered for adequately, then with hydraulic control of the sludge age, plant control is simplified and, on small scale plants, may even do away with the requirements for solids and SVI tests except at long intervals. Hydraulic control of sludge age makes the Sludge Loading Rate, SLR, (or Substrate Utilization Rate, SUR), redundant and introduces an entirely different attitude to process control.

11. SELECTION OF SLUDGE AGE

Selection of the sludge age is the most fundamental and important decision in the design of an activated sludge process. The sludge age of a plant depends on many factors *inter alia*, stability of the process, sludge settleability, whether or not the waste sludge should be suitable for direct discharge to drying beds, and most important of all, the quality of effluent required i.e. is COD removal only acceptable, must be effluent be nitrified, is nitrogen and phosphorus removal required. Although the discussion below will be aimed at South African conditions, where nitrification is obligatory and stringent standards for the effluent phosphate concentration have been promulgated, other situations will be discussed briefly so as to place South African activated sludge process technology in perspective with other technologies, for example those of Europe and North America.

11.1 Short sludge ages (1-5 days)

11.1.1 Conventional plants:

These plants are operated in the conventional con-

^{*}For a review of the development of hydraulic control of sludge age see Ekama and Marais (1978).

^{**}It should be noted that in a gravity thickener for waste sludge from a nutrient removal plant, considerable concentrations of phosphorus may be released in the thickener, which, when discharged to the activated sludge process will be counter-productive for excess biological P removal. The thickener supernatant may have to be treated chemically to precipitate the P before discharge to the activated sludge process (see Section 8 above).

figuration i.e. a semi plug flow configuration, but modified systems such as contact stabilization, step aeration, step feed and others are also implemented.

Short sludge age plants are extensively used in Europe and North America but are virtually unknown in South Africa principally because the promulgated standards for effluents cannot be satisfied, i.e. that of obligatory nitrification.

Short sludge age plants are for the purpose of COD removal only, towards which objective sludge ages of 1 to 3 days are sufficient. BOD or COD reductions, for the conventional system range from 75 to 90%; the actual removal achieved depends on the influent, the state of the sludge and the efficiency of the settling tanks. Predatory activity of the micro-organisms in the sludge is relatively low which causes turbidity and high effluent COD.

Nitrification should be absent, or the plant can be operated to prevent nitrification; this however may be difficult to achieve with high water temperatures in the tropics.

Advantages with low sludge age plants are: small process volumes for unit input COD (Figure 4.3) with corresponding low carbonaceous oxygen demand (see Figure 4.2); nitrification is unlikely, giving a saving in the nitrification oxygen demand (see Figure 6.1); oxygen demand under cyclic flow and load tends to be heavily damped making control of the oxygen easier than with longer sludge age plants. Disadvantages are: the mass of sludge produced by unit COD input is high (Figure 4.4) and the active fraction of the sludge is high (Figure 4.2). The sludge treatment facilities accordingly constitute an appreciable fraction of the treatment costs although the running costs can be reduced by energy recovery from anaerobic digestion of the sludge (see below); the process tends to be unstable and sensitive to mechanical breakdown.

Because of the relatively low oxygen demand and the high sludge production per unit input COD it is possible to make these high rate processes near self sufficient with respect to energy requirements – sufficient methane gas can be produced by anaerobic digestion of the waste sludge to generate electricity for oxygen supply and process operation.

11.1.2 Aerated lagoons:

In aerated lagoons, (as opposed to aerated oxidation ponds which supplement the oxygenation by algae) the oxygen requirement in the process is supplied wholly by aerators, usually floating mechanical ones. There are essentially two types of aerated lagoons, suspension mixed and facultative.

Suspension mixed aerated lagoons have sufficient energy input per unit volume by the aeration equipment to keep the sludge in suspension. In facultative lagoons this energy input is insufficient and settlement takes place on the lagoon floor, to form a sludge layer which decomposes anaerobically, as in an oxidation pond.

Kinetically, suspension mixed lagoons are members of the activated sludge process, the only difference being in that the former is a flow through system i.e. the sludge age equals the nominal hydraulic retention time, whereas the latter has a sludge recycle from the settling

tank making the sludge age longer than the hydraulic retention time. Accordingly the volume of the aerated lagoon per unit input COD is large whereas it is small for the conventional short sludge plant. (For a sludge age of, say, 1 day the aerated lagoon will have an hydraulic retention time of one day whereas the short sludge age conventional plant (with a one day sludge age) will have a hydraulic retention time of about 1/8 of a day.

The effluent from a suspension mixed aerated lagoon has the same constitution as the mixed liquor in the basin. The COD removed from the system via the oxygen demand is relatively small so that the COD in the effluent is quite unacceptable for discharge to receiving waters. In fact the principal objective of all short age plants is to act as biological flocculators, to physically change the influent COD, soluble and particulate, to a form that allows effective liquid-solid separation. In the conventional short sludge age plant the waste sludge is transferred to the sludge treatment facility; in the aerated lagoon usually a second pond is provided i.e. an oxidation pond or a facultative aerated lagoon, to allow the now readily settleable particulate material to settle as a sludge layer and thereby produce a relatively solids-free effluent.

Aerated lagoons find application principally in industrial waste treatment and seasonal waters. Design considerations are not dealt with in this manual; these are discussed by Marais and Ekama (1980) and reference should be made to that report.

11.2 Intermediate sludge ages (10-15 days)

Where nitrification is obligatory, the sludge ages required are 5 to 8 times longer than those for COD removal only, depending on the temperature. In the temperate regions where water temperatures may fall below 12°C to 14°C the sludge age is unlikely to be less than 10 to 15 days. In this range of sludge age, the effluent COD concentration no longer plays a role in the design because for sludge ages longer than about 4 days, the effluent COD or BOD concentrations remain approximately constant (see Section 3.4 above). The effluent ammonia concentration also plays a minor role in design because nitrification kinetics are such that once nitrification is achieved, it is virtually complete; even though the effluent standards may require an effluent ammonia concentration of less than, say, 10 mgN/l, once nitrification takes place the concentration is likely to be less than 4 mgN/l. Consequently for nitrification, the sludge age of the process is fixed completely by the requirement for nitrification. (The method for calculating the minimum sludge age for nitrification is given in Chapter 5, Section 3.)

With low alkalinity wastewaters (like those encountered in the Western and Southern Cape), nitrification can cause a significant reduction in effluent pH, often as low as 5. This not only causes problems with the nitrification process itself in that only partial nitrification is achieved, with the likelihood of non-compliance of the 10 mgN/l effluent ammonia standard (see Chapter 5, Section 4.3), but it also tends to favour the development of poor settling sludges and to produce aggressive effluents that can do considerable damage to concrete surfaces. To reduce these problems (and to

derive other advantages cited later in this section) the policy of deliberate biological denitrification has evolved. However, once biological denitrification is incorporated in the process, sludge ages longer than 10 to 15 days are required and the process falls into the long sludge age category.

Comparing intermediate sludge age plants with high rate plants, the oxygen demand per kgCOD (including nitrification) is doubled (see Figure 6.1), the process volume is 3 to 4 times larger (see Figure 4.3), the daily sludge mass wasted is reduced by 40% (see Figure 4.4) and active fraction is much lower (see Figure 4.2). Intermediate sludge age plants are much more stable than high rate plants, requiring less sophisticated control techniques or operator intervention thereby making these plants more suitable for general application.

At intermediate sludge ages, the active fraction of the waste sludge is still too high for direct discharge to drying beds. Consequently some form of waste sludge stabilization would need to be incorporated in the treatment works, i.e. either aerobic or anaerobic digestion. The former has the advantage of ease of operation but the disadvantage of energy costs for oxygen supply; the latter has the advantage of energy production but the disadvantage of complexity of operation. Even with energy recovery by anaerobic digestion of waste sludge, because of the low mass of sludge wasted from the activated sludge plant and high oxygen demand per kgCOD load, energy self-sufficiency at intermediate sludge ages is impossible. However, on large plants (approximately 500 000 person equivalent) where technical supervision and operator expertise are of a high level, energy cost can be reduced by gas production from anaerobic digesters and probably can be justified economically, particularly if energy costs continue to increase as they have over the past decade.

In nitrifying aerobic activated sludge plants, there is always the possibility that denitrification can take place in the secondary settling tank. This problem is exacerbated by the process control procedure of abstracting the waste sludge from the settling tank underflow (see Section 10 above). This procedure is used to benefit from the thickening function of the settling tank: the lower the underflow recycle ratio, the higher the underflow concentration, the lower the volume of waste sludge. However, low underflow recycle ratios result in large sludge accumulations in the secondary settling tank (see Figure 4.8). Hence the sludge might be retained in the settling tank for a long period of time and this is likely to lead to denitrification. The degree of denitrification is increased as:

- (i) the residence time in the settling tank increases – this depends on the recycle ratio and peak flow conditions;
- (ii) the active fraction of the sludge increases, i.e. is greater at shorter sludge ages, Figure 4.2;
- (iii) the temperature increases; and
- (iv) the mass of biodegradable COD adsorbed on the organism mass increases (van Haandel, Ekama and

Marais, 1981) – this is generally greatest at the peak load condition.

If denitrification in the settling tank takes place to a sufficient degree, attachment of sludge particles to the nitrogen gas bubbles causes buoying of the particles to the surface. The flotation effect can cause considerable loss of sludge via the settling tank overflow. This behaviour has been observed, for example, at Bellville, Cape Town, where sludge losses have been observed during that latter part of summer when the water temperatures are around 22°C. Sludge loss occurs daily in the afternoon after the peak flow and load when both the sludge accumulation in the settling tank and stored COD adsorbed on the sludge mass are at a maximum. Although the underflow recycle ratio is quite high ($s = 2$) sufficient sludge still accumulates to make denitrification significant at high temperatures.

The above discussion on experience with nitrifying activated sludge plants points to the following conclusions:

The secondary settling tank no longer should serve the dual purpose of solid-liquid separation and thickening. The sludge residence time needs to be minimized, to reduce denitrification, by having a high underflow recycle ratio, of 1 to 2:1, but this will result in inadequate thickening of the waste sludge withdrawn from the underflow. The thickening aspect can be solved by implementing hydraulic control of the sludge age, by providing a small independent satellite settling tank for gravity thickening of the waste sludge (the supernatant being returned to the reactor so that inadequate clarification due to denitrification flotation does not affect effluent quality) or by replacing the gravity satellite tank with a dissolved air flotation unit.* These modifications, however, ameliorate the effects on the secondary settling tank but do not positively remove the root cause, i.e. the high nitrate concentration in the mixed liquor. This can be done only by including denitrification in the process, which then shifts the process to that of the long sludge age type.

11.3 Long sludge ages (20 days or more)

11.3.1 Aerobic plants:

Long sludge age aerobic plants are usually called extended aeration plants. The sludge age is chosen to be of such a length that the active mass fraction of the waste sludge is sufficiently reduced to allow discharge directly to sludge drying beds after thickening. What this sludge age should be will depend to a degree on the climatic conditions i.e. whether the sludge can be dried before it starts smelling, but probably exceeds 30 to 40 days – research into the sludge ages necessary under South African conditions is necessary.

Compared to intermediate sludge age plants the total oxygen demand (i.e. carbonaceous plus nitrification) is about equivalent (Figure 6.1); the process

*Thickening by dissolved air flotation now is both technically and economically feasible, Bratby and Marais (1976, 1977).

volume requirement is 50 to 60% higher and the active fraction 50% lower. As nitrification always occurs, problems with rising sludge in the settling tank are to be expected. Control of the sludge age is best via the hydraulic sludge age control method, so that high recycles from the secondary settling tank can be imposed to reduce rising sludge in the tank, as described under intermediate sludge age processes above. Problems of low pH when treating low Alkalinity wastewater are to be expected.

Extended aeration plants are very stable in operation and require probably less supervision than any other activated sludge process. Although the volume requirements and the oxygen demand per unit input COD is large, the relative ease of operation makes this process the indicated one for small communities. However, it must be remembered that this process will discharge a low COD effluent but with high nitrate and phosphate concentrations. The nitrate concentration can be drastically reduced and low pH effluent prevented by incorporating an anoxic zone in the process without basically disturbing the ease of operation. This modification always should be kept in mind when considering extended aeration as a possible process.

Processes that can be operated in the extended aeration modes are the completely mixed single reactor, the Orbal multi-channel, oxidation ditch and Carrousel processes.

11.3.2 Anoxic-aerobic process

Once sludge ages in the range greater than 20 days are accepted it is possible usually to incorporate denitrification in the process without upsetting stability in nitrification. The reactor is suitably subdivided into unaerated (anoxic) and aerated zones in a variety of ways, the denitrification taking place in the unaerated zones and the nitrification in the aerated zones to give the so-called nitrification-denitrification process. The advantages are substantial: (1) Nitrate concentration in the effluent is reduced, the nitrate utilized in the unaerated zones lowering the oxygen demand of the process accordingly; with complete denitrification the total oxygen demand can be reduced by approximately 15 to 20% compared with the requirements for a nitrifying process. (2) Loss of sludge in the overflow from the settling tank due to rising sludge is largely eliminated. (3) Alkalinity recovery with denitrification causes pH instability with low influent alkalinity wastewaters to be very unlikely. Normally the sludge ages in nitrification-denitrification plants are insufficient to allow direct discharge of the waste sludge to the drying beds.

Process configurations of anoxic-aerobic systems are the Modified Ludzack-Ettinger process in which a primary anoxic reactor is provided, the Bardenpho which incorporates both primary and secondary anoxic reactors, the Orbal in which anoxic zones are created by reducing the aeration energy input in channels, Carrousel and oxidation ditch in which the aeration at the aerators is controlled to induce anoxic operation in the same reactor, or in two interlinking parallel reactors.

Incorporation of biological denitrification in a process imposes additional constraints on the design. It requires the selection of a process sludge age *and* an

unaerated mass fraction, (to achieve the required denitrification or the highest denitrification possible), yet ensuring efficient nitrification under all expected conditions. (See Chapter 5, Section 4.4, and Chapter 6, Section 6).

11.3.3 Anaerobic-anoxic-aerobic systems

These plants are needed where biological removal of both nitrogen and phosphorus is required and are called biological excess P removal plants. The phosphorus removal propensity is created by providing a zone in the process that receives the influent wastewater but not oxygen and nitrate (via the recycles of mixed liquor). The incorporation of this zone imposes further constraints on the design of the system: A sludge age must be selected that will ensure efficient nitrification at all times, provide for anoxic zones that will give complete or maximal removal of nitrate and an anaerobic zone that receives the influent flow and recycles of mixed liquor not containing nitrate generated in the system. The critical aspects of the design are the assurance of no discharge of nitrate to the anaerobic zone. The success of the design is sensitive to a number of factors: the influent wastewater characteristics, such as the readily biodegradable COD concentration, the TKN/COD and P/COD ratios, the maximum specific growth rate of the nitrifiers, maximum and minimum temperatures, severity of cyclic load and flow, control of aeration and control of nitrate and dissolved oxygen in some of the recycles.

Aeration control is a particularly vexing problem under cyclic load and flow conditions because the process is affected by too high or too low oxygen concentrations in the aerobic zone. Too high concentrations cause oxygen to be recycled to the anoxic zone (and the anaerobic zone in the 5 stage Phoredox process), thereby reducing the potential for P and N removal; too low oxygen concentrations cause the nitrification efficiency to decline and poor settling sludges to develop.

Oxygen control setups that have been developed still require considerable operational attention, particularly the oxygen probes, and at present their installation in smaller plants is to be questioned. These difficulties have prompted research into an alternative solution – load and flow equalization. In this approach an equalization tank is provided upstream of the plant. The effluent flow from the tank is controlled in such a manner that the cyclic fluctuations in the flow *and* load are damped to very small values. The control is via a microcomputer that calculates the discharge flow setting. The essential inputs to the computer are the depth of liquor in the equalization tank and the flow rate out of the tank; both these measurements can be readily monitored automatically and operated reliably over long periods of time. The microcomputer also has reached a stage of development that ensures its long term trouble-free service. The equalization approach with the microcomputer control, has been tested at Goudkoppies plant, Johannesburg and shows great potential for reducing aeration and other control problems in nutrient removal plants. Detailed information on equalization tanks is available in a report by Dold, Buhr and Marais (1982). Design details of the P removal

processes, considered in this monograph, are given in Chapter 7.

It should be noted that the waste sludge from a biological excess P removal activated sludge process contains high concentrations of P. If the sludge is allowed to become anaerobic or is anaerobically digested much of the phosphorus removed from the wastewater will be released from the sludge mass to the bulk liquid; consequently seepage from drying beds may have to be chemically treated to precipitate the phosphorus before recycling to the plant. Failure to recognize this behaviour of phosphorus-rich biological sludges may lead to inadvertent re-contamination of the effluent with high concentrations of phosphorus (see Section 8 above).

Examples of N and P removal processes are the multi-reactor Phoredox and UCT processes in which the anaerobic, anoxic and aerobic conditions are induced in separate reactors interlinked by recycle flows. Other removal systems have been proposed, for example that by Kerdachi and Roberts (1982) in which these conditions are imposed within the same reactor.

Where P removal without obligatory nitrogen removal is required the sludge age can be selected to prevent nitrification and in the multi-reactor type plant theoretically only anaerobic and aerobic reactors are needed, i.e. no anoxic reactor. The sludge age is reduced substantially, probably to less than eight days, so that the reactor volume per input COD is much smaller than where nitrification-denitrification is a requirement. However, even with these short sludge ages it would be wise, always, to make provision for denitrification of any nitrate generated and recycled to the anaerobic zone. This provision is of particular relevance in warm climates or where the maximum specific growth rate of the nitrifiers is very high so that nitrification can occur at aerobic sludge ages of, say, 2 days. These short sludge age P removal plants are not considered in this monograph.

is demonstrated with numerical examples. Assuming constant flow and load conditions, calculations are presented showing how estimates of the process volume requirements, average daily carbonaceous oxygen demand and daily sludge production are obtained for the treatment of raw or settled wastewaters based on the wastewater characteristics.

12.1 Influent Wastewater Characteristics

The daily average characteristics for a particular raw wastewater are given in Table 4.3. In the design example it is assumed that primary sedimentation tanks remove 40% of the total COD load, 15% of the TKN and total P and 10% of the readily biodegradable COD. The settled wastewater characteristics are also given in Table 4.3. It should be noted that the *soluble concentrations* in the settled wastewater should be approximately equal to those in the raw wastewater. For example, from Table 4.3 the soluble unbiodegradable COD concentration in the raw wastewater is 0,05 mg COD/mg COD, i.e. $0,05 \cdot 600 = 30$ mg COD/l (Eq. 2.3). As the soluble unbiodegradable constituents are not affected by primary sedimentation, the unbiodegradable soluble COD *fraction* of the settled wastewater accordingly will be approximately $30/360 = 0,08$ mg COD/mg COD. Similarly, the readily biodegradable COD fraction (f_{bs}) of the raw wastewater is given as 0,24. This is equivalent to a concentration of $0,24 (1 - f_{up} - f_{us}) S_{ti} = 0,24(0,82)600 = 118$ mg COD/l (see Eqs. 2.8 and 2.9). Primary sedimentation is assumed to remove 10% of the readily biodegradable COD (due to biodegradation), the f_{bs} *fraction* in the settled wastewater is given by $\{(1,0 - 0,10)118\} / \{(1 - 0,08 - 0,04)360\} = 0,33$ (see Eqs. 2.8 and 2.9). The examples show that care must be taken that settled wastewater characteristics are not fixed at values that are inconsistent with the raw wastewater characteristics and primary settling tank behaviour.

12. DESIGN EXAMPLE

In this section, the procedure for design of organic material (COD) degradation in purely aerobic processes

12.2 Temperature effects

From Table 4.1, the only constant in the organic material degradation theory that is affected by

TABLE 4.3 RAW AND SETTLED WASTEWATER CHARACTERISTICS (SEE ALSO TABLES 5.2 AND 6.1)

Parameter	Symbol	Value		Units
		Raw	Settled	
Influent COD concentration	S_{ti}	600*	360*	mgCOD/l
Influent TKN concentration	N_{ti}	48*	41*	mgN/l
Influent P concentration	P_{ti}	10*	8,5*	mgP/l
TKN/COD ratio	—	0,080	0,114	mgN/mgCOD
P/COD ratio	—	0,017	0,024	mgP/mgCOD
Unbiodegradable soluble COD	f_{us}	0,05	0,08	mgCOD/mgCOD
Unbiodegradable particulate COD	f_{up}	0,13	0,04	mgCOD/mgCOD
MLVSS/MLSS ratio of sludge produced	f_i	0,75	0,83	mgVSS/mgTSS
Temperature — Summer	T_{max}	22	22	°C
— Winter	T_{min}	14	14	°C
pH of wastewater	—	7,5	7,5	—
Influent flow	Q	13,33	13,33	ML/d

*Influent concentrations appropriately weighted to take account of daily cyclic flow variations

temperature is the specific endogenous respiration rate, b_h . This rate reduces by about 3% every 1°C drop in temperature – from Eq. 4.5, (given in Table 4.1), the rate at 14°C is 0,20/d and at 22°C is 0,25/d. The effect of the reduction in the rate with decrease in temperature is that at reduced temperatures the daily sludge production is marginally increased and the average carbonaceous oxygen demand is marginally decreased – the differences in sludge production and oxygen demand are less than 5% for a 6°C change in temperature. Consequently, the average carbonaceous oxygen demand should be calculated at the maximum temperature and the process volume and sludge production at the minimum temperature in order to find the maximum values of these parameters.

12.3 Calculations for organic material degradation

The example will demonstrate the effect of temperature and sludge age on (i) the mass of sludge in the process, (ii) the average daily carbonaceous oxygen demand, (iii) the active fraction of the sludge and (iv) mass of sludge wasted daily. These four parameters will be calculated for the raw and settled wastewater at 14°C and 22°C for sludge ages ranging from 5 to 30 days. The values of the constant Y_h , f_{cv} , f and b_h (appropriately adjusted for temperature) are given in Table 4.1.

$$\text{Mass of COD treated per day} = M(S_{ti}) = Q \cdot S_{ti} \quad (4.6)$$

$$\text{Mass of biodegradable COD} = M(S_{bi}) = (1 - f_{up} - f_{us})M(S_{ti}) \quad (4.7c)$$

$$\text{Mass of unbiodegradable particulate solids} \\ M(X_{ii}) = M(S_{ti})f_{up}/f_{cv} \quad (4.8c)$$

Hence for raw wastewater:

$$M(S_{ti}) = 13,33 \cdot 10^6 \cdot 600 \text{ mg COD/d} = 8000 \text{ kg COD/d}$$

$$f_{us} = 0,05 \text{ mg COD/mg COD}$$

$$f_{up} = 0,13 \text{ mg COD/mg COD}$$

$$M(S_{bi}) = (1 - 0,05 - 0,13)8000 = 6560 \text{ kg COD/d}$$

$$M(X_{ii}) = 8000 \cdot 0,13/1,48 = 702,7 \text{ kg VSS/d}$$

and for settled sewage:

$$M(S_{ti}) = 60\% \text{ of } 8000 = 4800 \text{ kg COD/d}$$

$$f_{us} = 0,08 \text{ mg COD/mg COD}$$

$$f_{up} = 0,04 \text{ mg COD/mg COD}$$

$$M(S_{bi}) = (1 - 0,08 - 0,04)4800 = 4224 \text{ kg COD/d}$$

$$M(X_{ii}) = 4800 \cdot 0,04/1,48 = 129,7 \text{ kg VSS/d}$$

From Eq. 4.13 the mass of volatile solids in the process is given by

$$M(X_v) = \frac{0,45 R_s M(S_{bi})}{(1 + b_{hT} R_s)} (1 + 0,2 b_{hT} R_s) + M(X_{ii}) R_s$$

$$= \left\{ \frac{2952 (1 + 0,2 b_{hT} R_s)}{(1 + b_{hT} R_s)} + 702,7 \right\} R_s \quad (4.29a)$$

for raw wastewater; and

$$M(X_v) = \left\{ \frac{1901 (1 + 0,2 b_{hT} R_s)}{(1 + b_{hT} R_s)} + 129,7 \right\} R_s \quad (4.29b)$$

for settled wastewater.

From Eq. 4.14 the mass of total suspended solids in the process is given by

$$M(X_t) = M(X_v)/0,75 \text{ for raw wastewater} \quad (4.30a)$$

$$= M(X_v)/0,83 \text{ for settled wastewater} \quad (4.30b)$$

From Eq. 4.15, the average daily carbonaceous oxygen demand is given by

$$M(O_c) = M(S_{bi}) \{ (1 - f_{cv} Y_h) + f_{cv} (1 - f) b_{hT} \frac{Y_h R_s}{(1 + b_{hT} R_s)} \} \\ = 6560 \{ (0,334) + 0,533 \frac{b_{hT} R_s}{(1 + b_{hT} R_s)} \} \text{ (kg O/d)} \quad (4.31a)$$

for raw wastewater; and

$$= 4224 \{ (0,334) + 0,533 \frac{b_{hT} R_s}{(1 + b_{hT} R_s)} \} \text{ (kg O/d)} \quad (4.31b)$$

for settled wastewater.

From Eq. 4.18, the active fraction with respect to the volatile suspended solids (f_{av}) is given by

$$f_{av} = 1 / \{ 1 + 0,2 b_{hT} R_s + 0,238 (1 + b_{hT} R_s) \} \\ = 1 / \{ 1,238 + 0,438 b_{hT} R_s \} \text{ for raw wastewater.} \quad (4.32a)$$

$$f_{av} = 1 / \{ 1 + 0,2 b_{hT} R_s + 0,068 (1 + b_{hT} R_s) \} \\ = 1 / \{ 1,068 + 0,268 b_{hT} R_s \} \text{ for settled wastewater} \quad (4.32b)$$

From Eq. 4.22, the mass of sludge (as total suspended solids) produced per day is given by

$$M(\Delta X_t) = \frac{8000}{0,75} \left\{ \frac{0,82 \cdot 0,45}{(1 + b_{hT} R_s)} (1 + 0,2 b_{hT} R_s) + \frac{0,13}{1,48} \right\} \\ = 10667 \left\{ \frac{0,369 (1 + 0,2 b_{hT} R_s)}{(1 + b_{hT} R_s)} + 0,09 \right\} \quad (4.33a)$$

for raw wastewater; and

$$M(\Delta X_t) = \frac{8000}{0,83} \left\{ \frac{0,88 \cdot 0,45}{(1 + b_{hT} R_s)} (1 + 0,2 b_{hT} R_s) + \frac{0,04}{1,48} \right\} \\ = 5783 \left\{ \frac{0,396 (1 + 0,2 b_{hT} R_s)}{(1 + b_{hT} R_s)} + 0,03 \right\} \quad (4.33b)$$

for settled wastewater.

Substituting the b_{hT} value for 14°C (i.e. 0,202/d) and 22°C (i.e. 0,254/d) into Eqs. 4.25 to 4.29 the parameters $M(X_v)$, $M(X_t)$, $M(O_c)$, f_{av} and $M(\Delta X_t)$ can be calculated for sludge ages 5 to 30 days. The results are shown plotted in Figure 4.9.

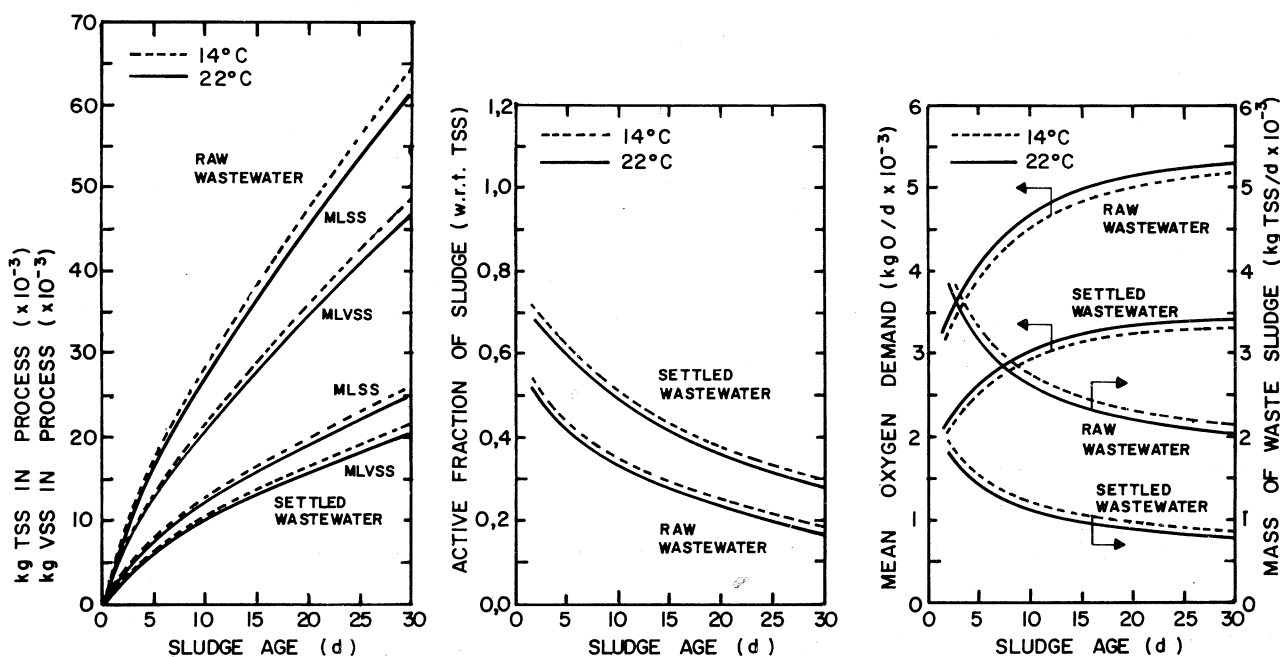


FIGURE 4.9: Mass of sludge (MLSS and MLVSS), active fraction (with respect to TSS), average carbonaceous oxygen demand and mass of sludge (as TSS) produced daily versus sludge age for raw and settled wastewaters at 14°C and 22°C. Wastewater characteristics given in Table 4.3.

Figure 4.9 confirms that the mass of sludge in the process (MLSS or MLVSS), the average carbonaceous oxygen demand and the active fraction (with respect to MLSS or MLVSS) are only marginally affected by temperature and for these parameters, insofar as design is concerned, are not really of consequence. However, the influent waste type i.e. raw or settled wastewater, has a significant effect; raw wastewater results in more sludge in the process, a higher oxygen demand and a lower active fraction of the sludge than settled wastewater. The difference in effects between raw and settled wastewater depends wholly on the efficiency of the primary settling tanks – the differences apparent in Figure 4.9 arise from a 40% COD removal in primary sedimentation. The greater the efficiency of primary sedimentation the greater the difference between the parameters shown in Figure 4.9.

From Eq. 4.16 for the same MLSS concentration, the process volume is proportional to the mass of sludge contained in it. Hence for the same MLSS concentration, the volume of the process treating settled wastewater will be only 40% of that treating raw wastewater at 25 days sludge age. Also, the settled wastewater plant requires only 64% of the oxygen the raw wastewater plant requires. However, the active fraction of the sludge in the settled wastewater plant is 34%, probably too high for direct discharge to drying beds, whereas that from the raw wastewater plant is 22%, probably sufficiently low for discharge to drying beds without further stabilization. Clearly the choice of treating settled sewage as against raw sewage requires

weighing the advantages and disadvantages of each against the other, i.e. settled sewage (smaller process volume, smaller oxygen demand and smaller secondary sludge production, but, with primary sludge production and its disposal) against raw sewage (greater process volume, higher oxygen demand and higher secondary sludge production, but no primary sludge production). The choice is made more difficult by the fact that higher biological removals of N and P are achieved with raw wastewater than with settled wastewaters (see Chapter 3, Section 3). These aspects are discussed in detail in Chapters 6 and 7 and will be demonstrated by extending the above design example to include biological nitrogen and phosphorus removal.

13. CONCLUSIONS

The calculations for the daily sludge production, daily average carbonaceous oxygen demand, process volume and nitrogen and phosphorus requirements for sludge production etc. were demonstrated for a raw and a settled sewage with selected wastewater characteristics for the raw sewage and assumed effects of primary settlement on the raw sewage characteristics. Selection of the numerical magnitudes of these characteristics was based on experience with five municipal waste flows so that the data should not be taken to apply generally. Whenever possible, the characteristics should be determined experimentally, or estimated by some semi-experimental procedure sup-

plemented by experience. Where design information is inadequate it will be necessary to select ranges of values for some parameters, ranges that appear to span across the true values, and to repeat the design a number of times using various combinations of the limiting values. In this way a picture will be built up of the sensitivity of the plant process response so that appropriate provisions can be made in the design. There are so many interactive effects that ample design provision in one aspect, for example, does not necessarily resolve that problem in terms of the whole design as it may create problems in other aspects.

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